

A

Colony area (pixels)

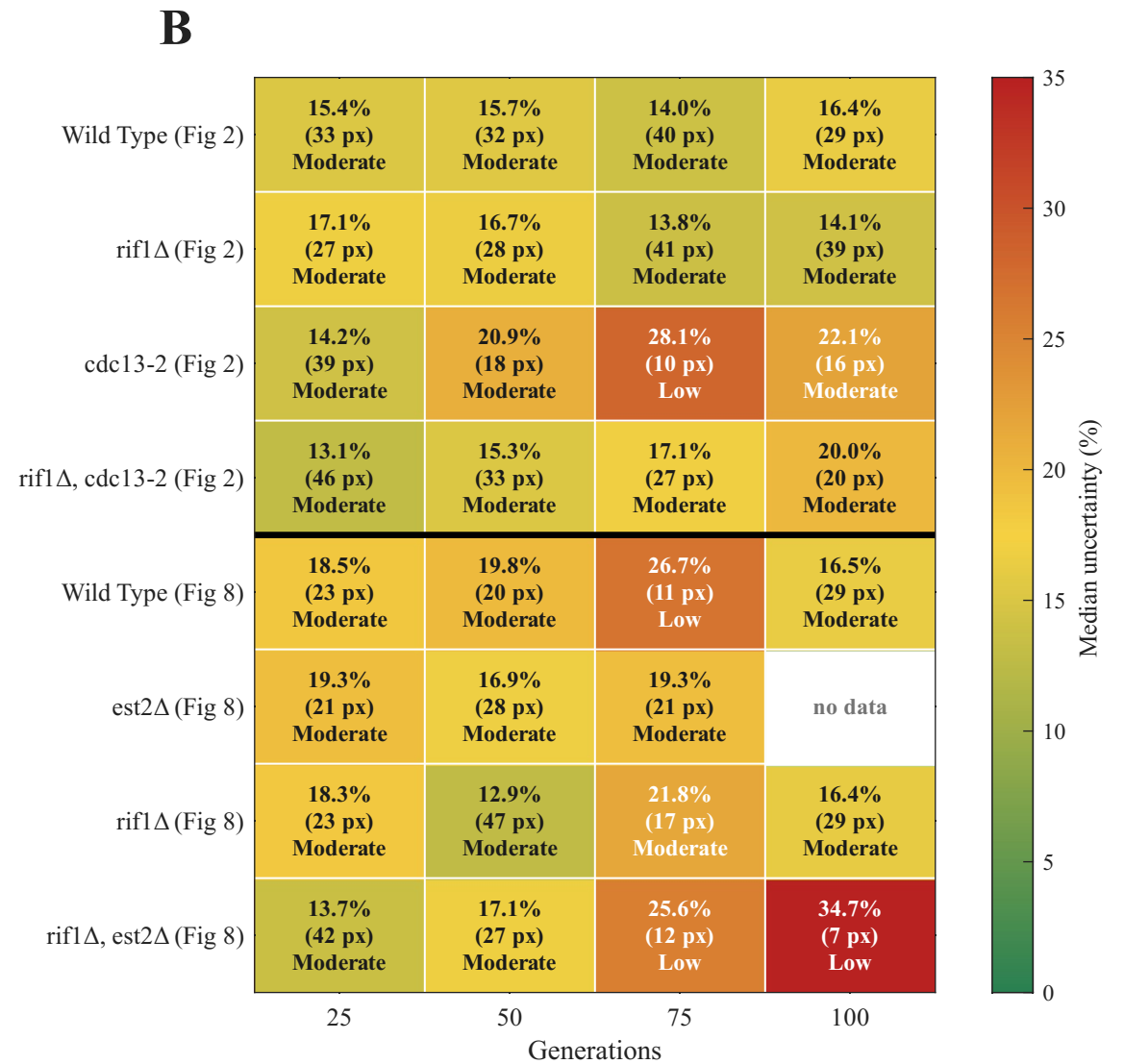
314 79 13 3

High Good Moderate Low Sub-pixel

median 16.7% (28 px)

Colonies (%)

Measurement uncertainty (% of colony radius)



(Figure 2)

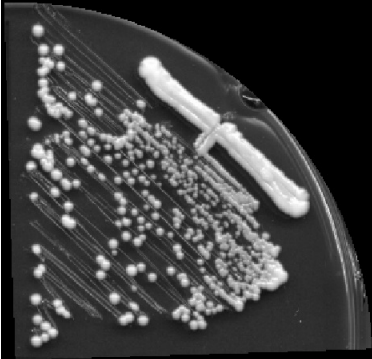
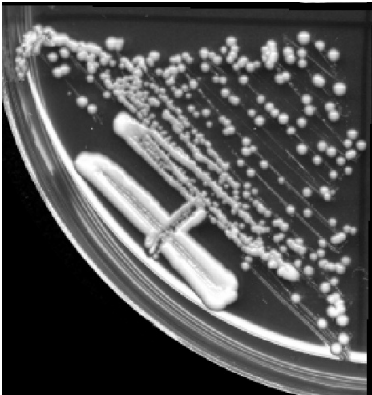
Wild Type

*rif1*Δ

cdc13-2

*rif1*Δ, *cdc13-2*

Passage 1



Colonies
Capture Ratio

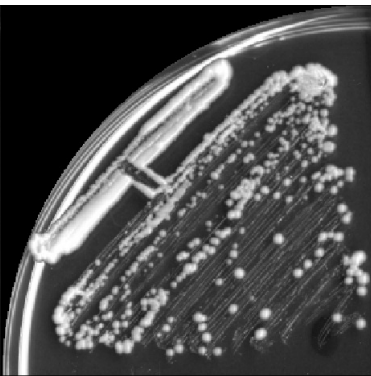
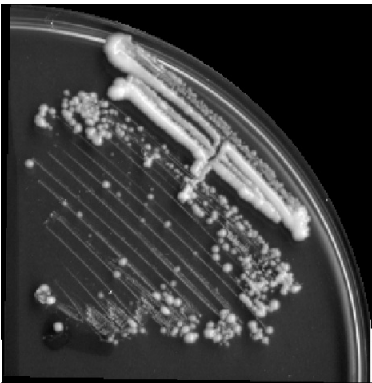
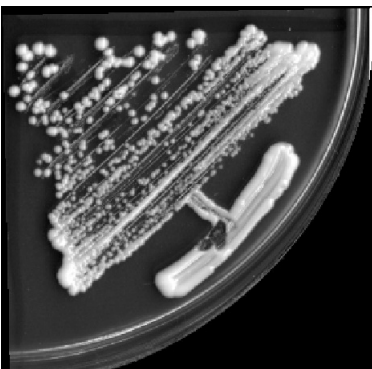
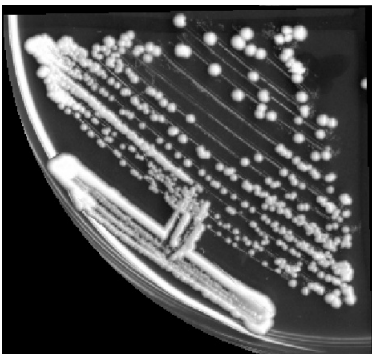
77
16.7%

57
7.5%

60
18.6%

34
10.9%

Passage 2



Colonies
Capture Ratio

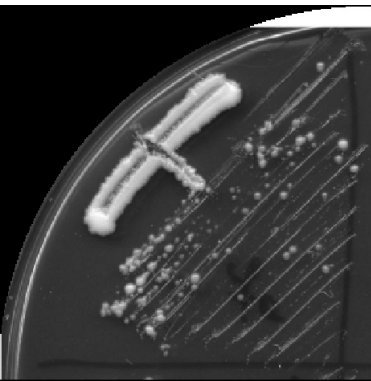
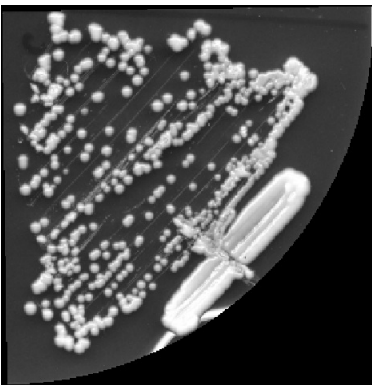
55
13.5%

27
5.4%

24
4.0%

48
10.0%

Passage 3



Colonies
Capture Ratio

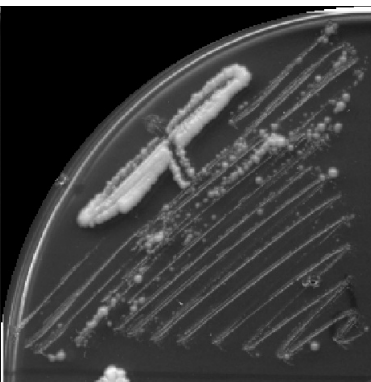
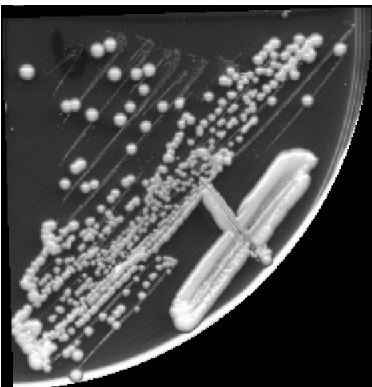
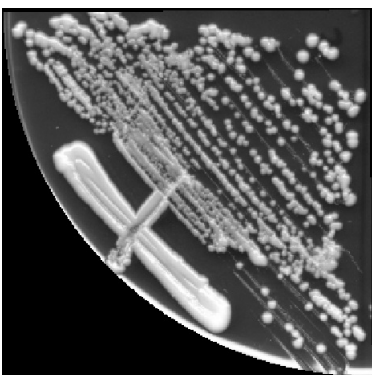
43
10.0%

71
16.9%

8
1.5%

24
8.9%

Passage 4



Colonies
Capture Ratio

56
8.4%

57
16.5%

29
8.8%

19
3.6%

(Figure 8)

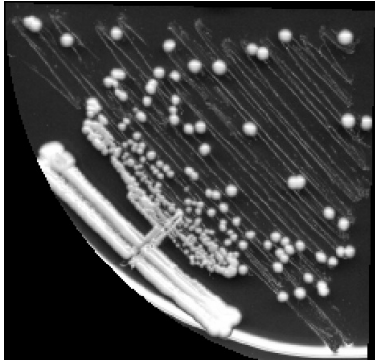
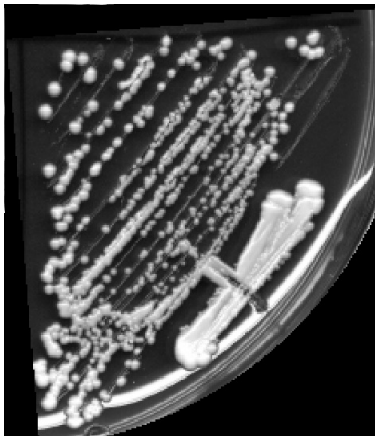
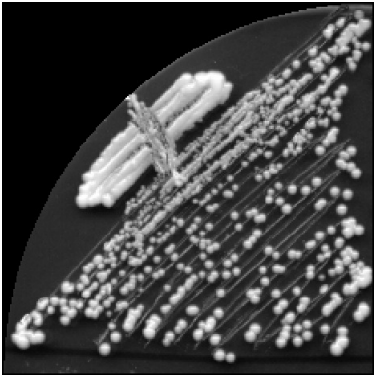
Wild Type

est2Δ

rif1Δ

rif1Δ, est2Δ

Passage 1



Colonies
Capture Ratio

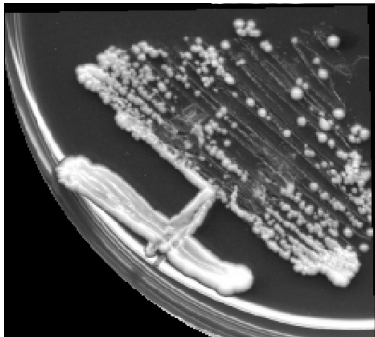
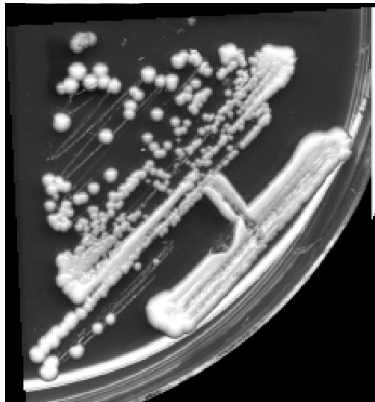
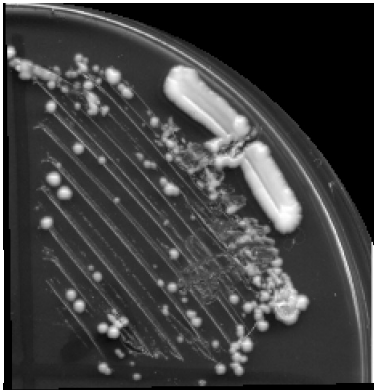
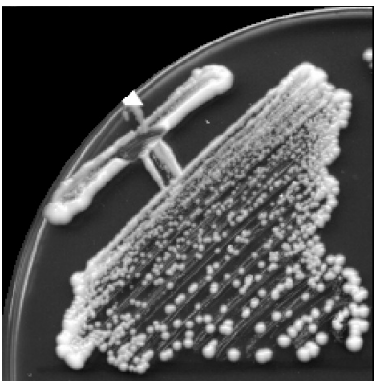
65
13.9%

43
10.4%

39
8.5%

44
22.3%

Passage 2



Colonies
Capture Ratio

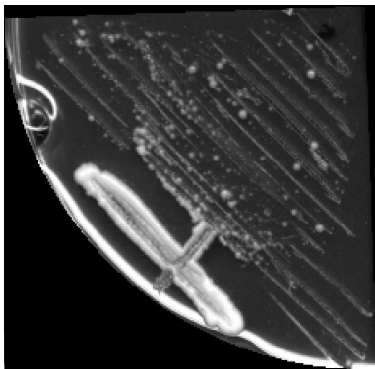
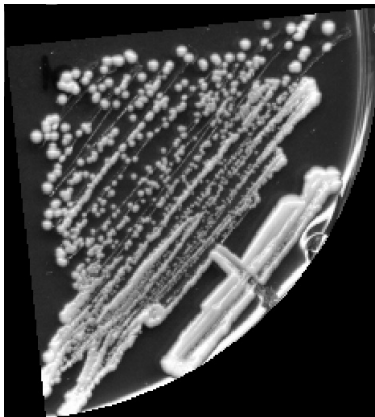
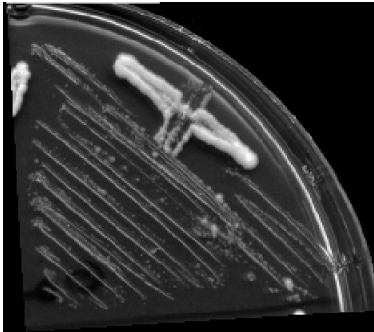
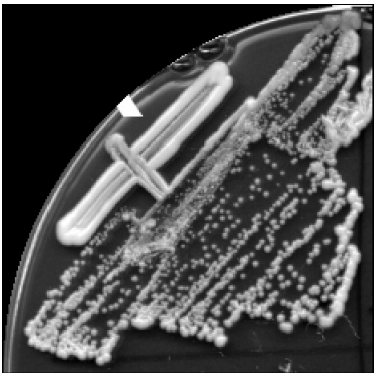
45
8.0%

24
9.9%

11
4.6%

19
4.9%

Passage 3



Colonies
Capture Ratio

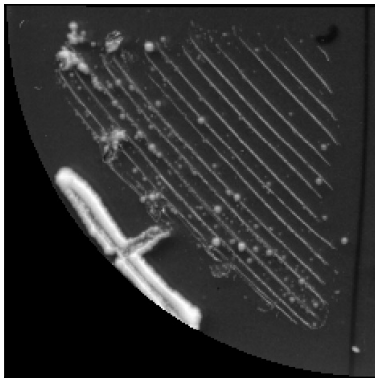
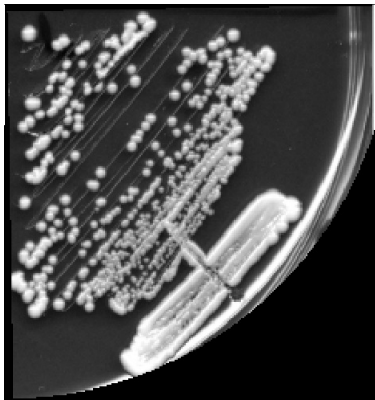
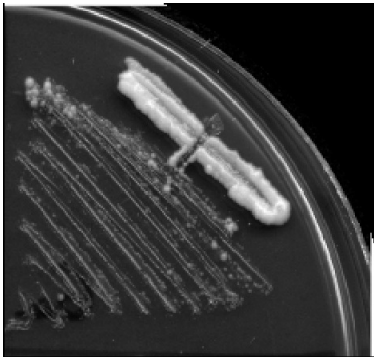
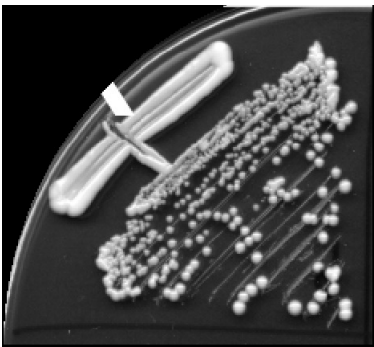
104
8.5%

5
1.7%

40
6.3%

15
3.5%

Passage 4



Colonies
Capture Ratio

57
16.4%

0
—

49
11.9%

38
6.6%

(Figure 2)

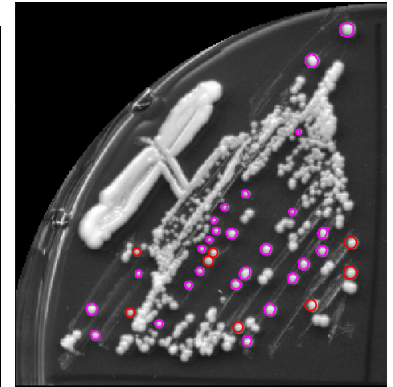
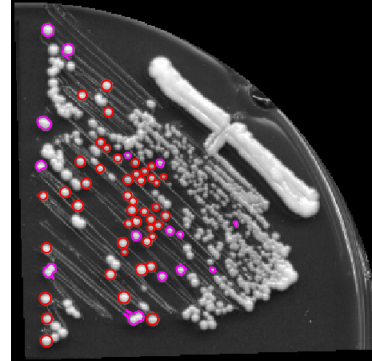
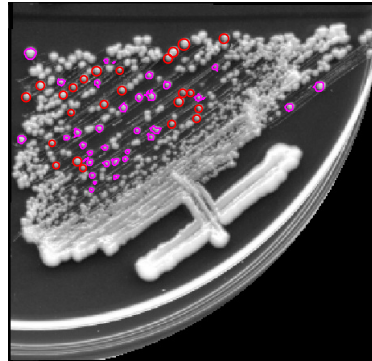
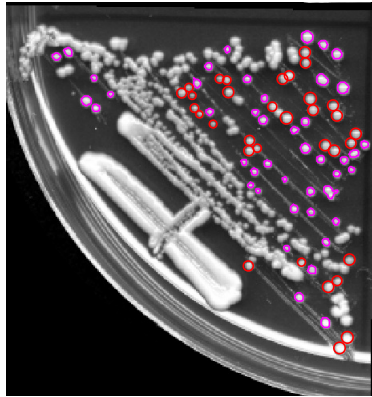
Wild Type

*rif1*Δ

cdc13-2

*rif1*Δ, *cdc13-2*

Passage 1



Colonies
Capture Ratio

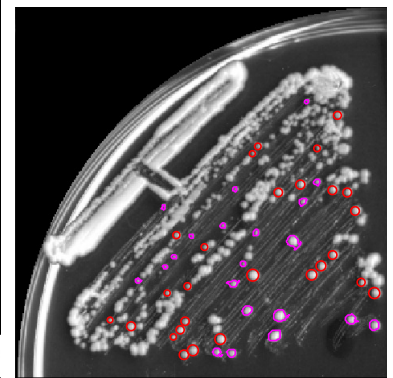
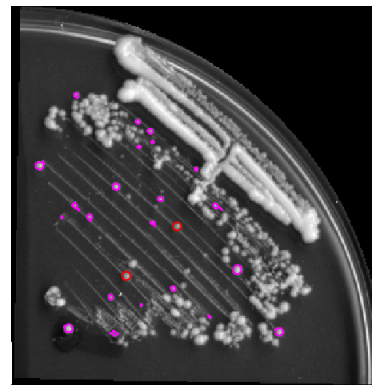
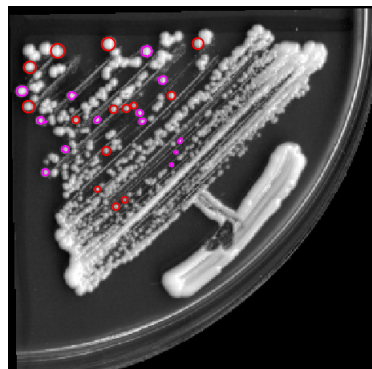
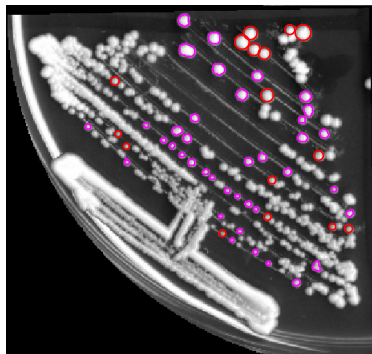
77
16.7%

57
7.5%

60
18.6%

34
10.9%

Passage 2



Colonies
Capture Ratio

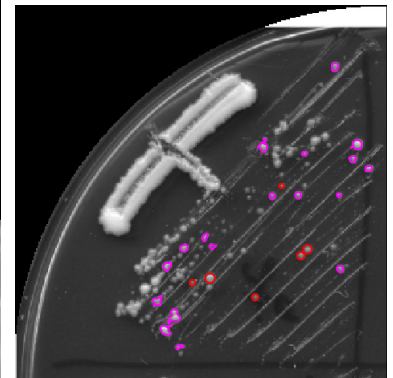
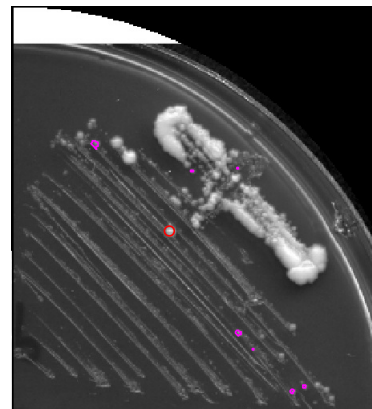
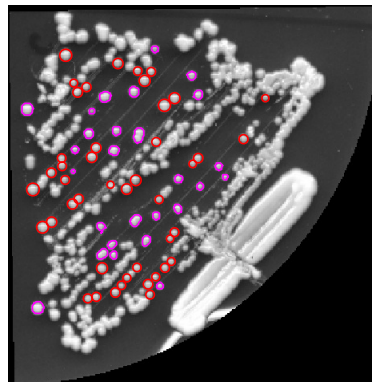
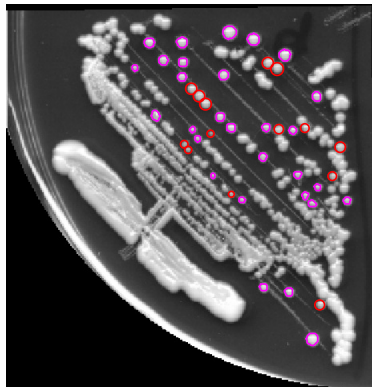
55
13.5%

27
5.4%

24
4.0%

48
10.0%

Passage 3



Colonies
Capture Ratio

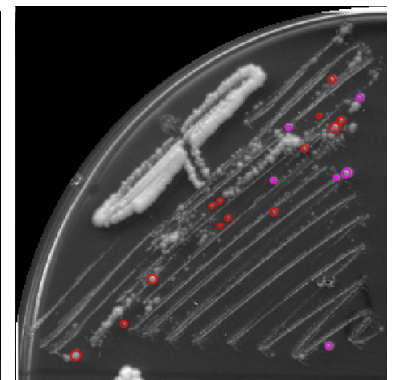
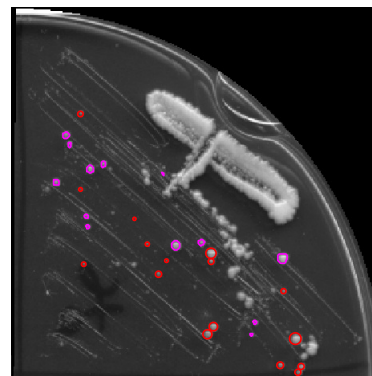
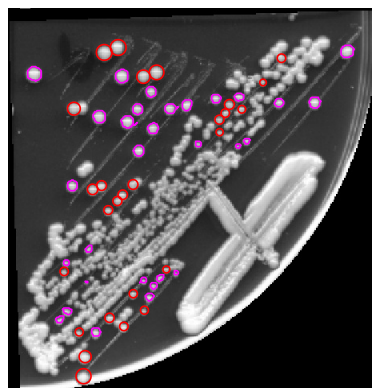
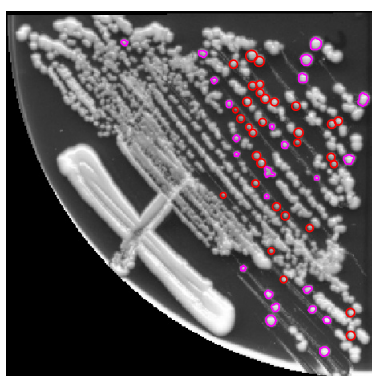
43
10.0%

71
16.9%

8
1.5%

24
8.9%

Passage 4



Colonies
Capture Ratio

56
8.4%

57
16.5%

29
8.8%

19
3.6%

Plate diameter: 90 mm

(Figure 8)

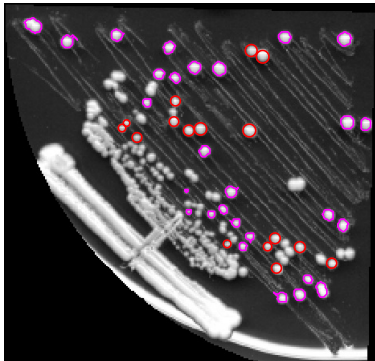
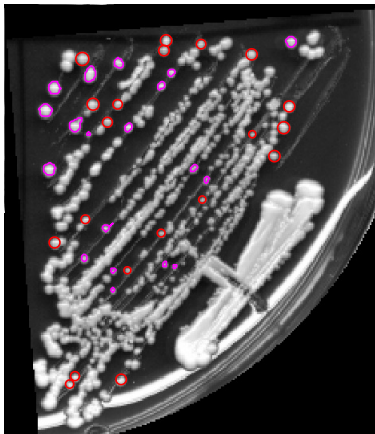
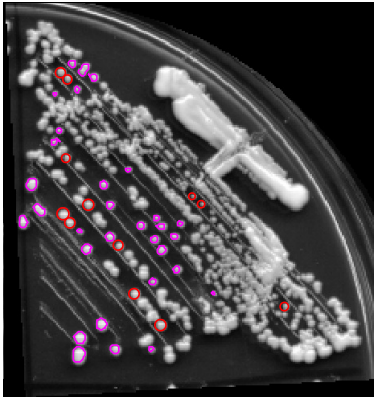
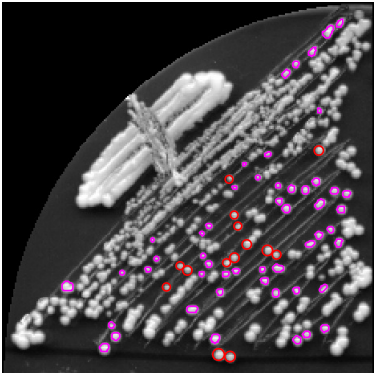
Wild Type

est2Δ

rif1Δ

rif1Δ, est2Δ

Passage 1



Colonies
Capture Ratio

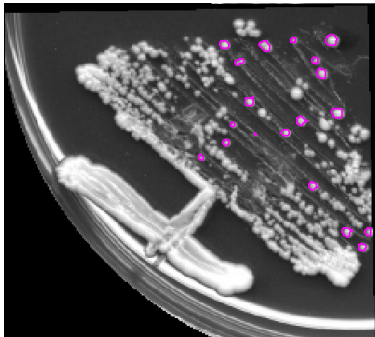
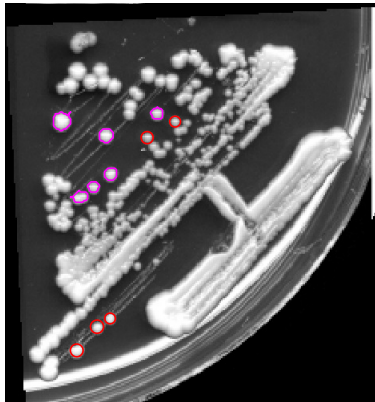
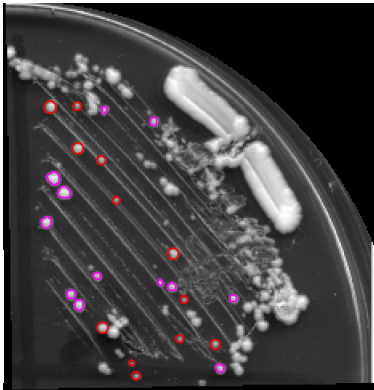
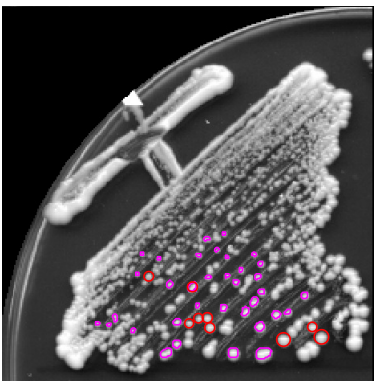
65
13.9%

43
10.4%

39
8.5%

44
22.3%

Passage 2



Colonies
Capture Ratio

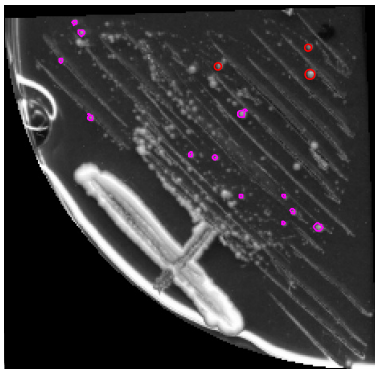
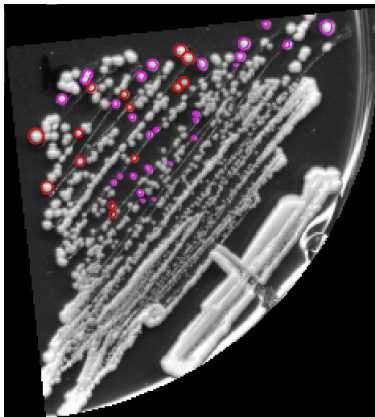
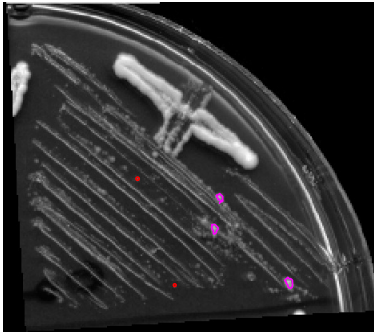
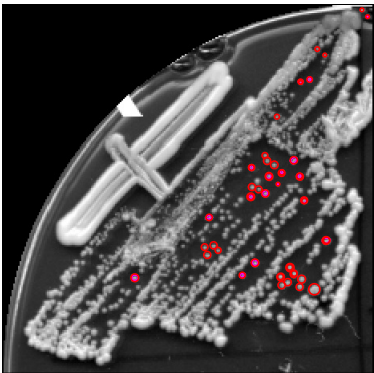
45
8.0%

24
9.9%

11
4.6%

19
4.9%

Passage 3



Colonies
Capture Ratio

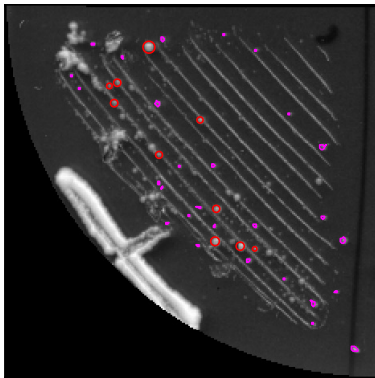
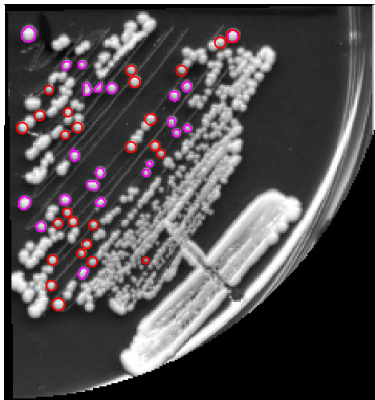
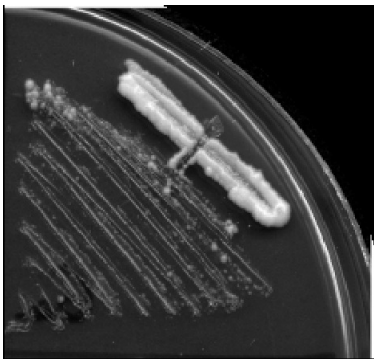
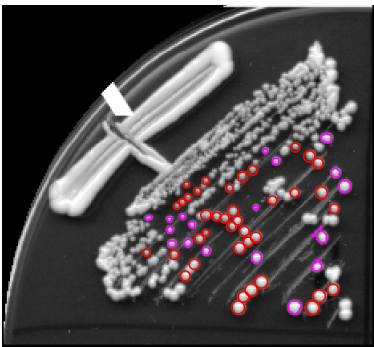
104
8.5%

5
1.7%

40
6.3%

15
3.5%

Passage 4



Colonies
Capture Ratio

57
16.4%

0
—

49
11.9%

38
6.6%

Plate diameter: 90 mm

Density Histograms (Figure 2)

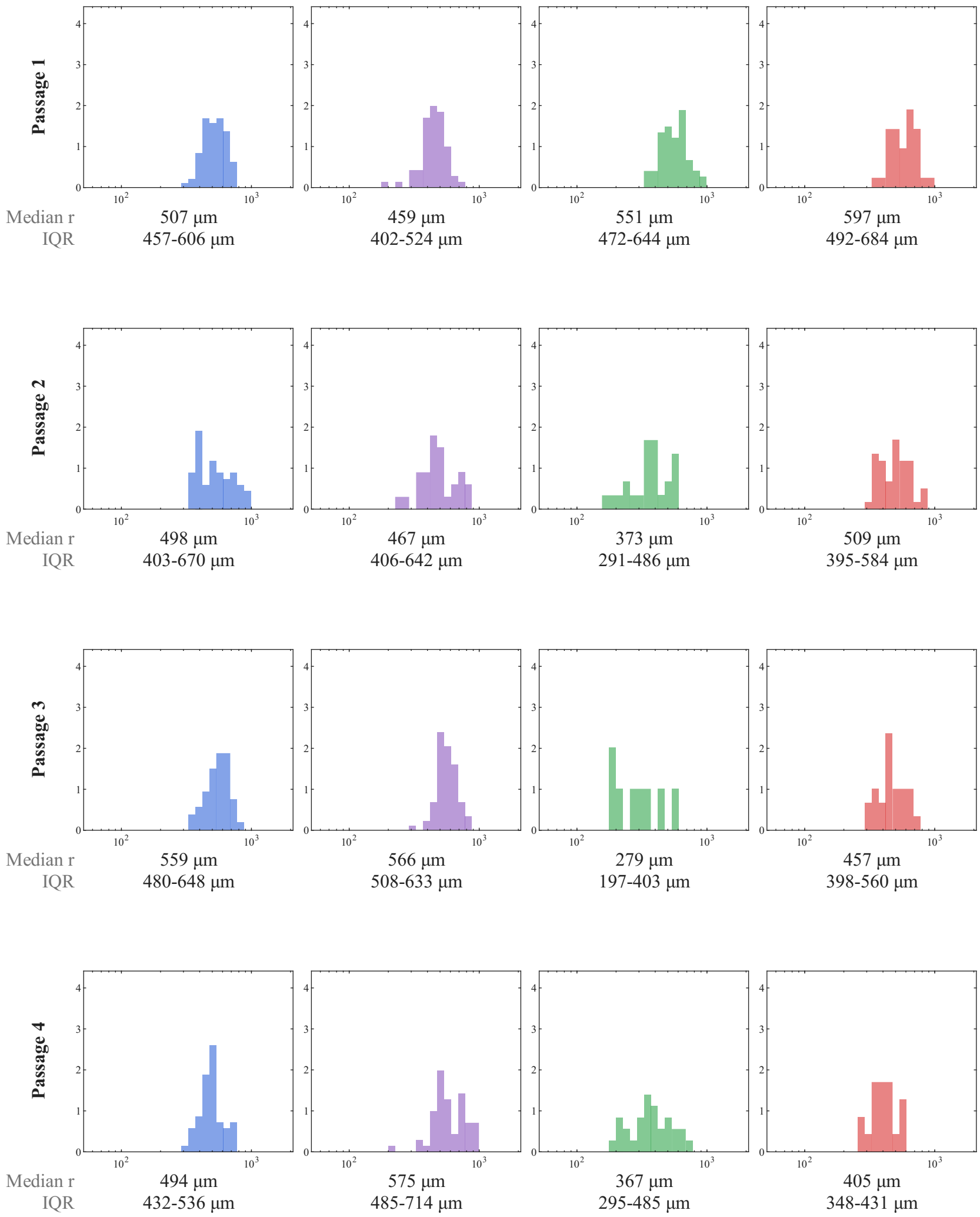
X axis: Colony Radius μm , Y axis: Density

Wild Type

rif1 Δ

cdc13-2

rif1 Δ , *cdc13-2*



Density Histograms (Figure 8)

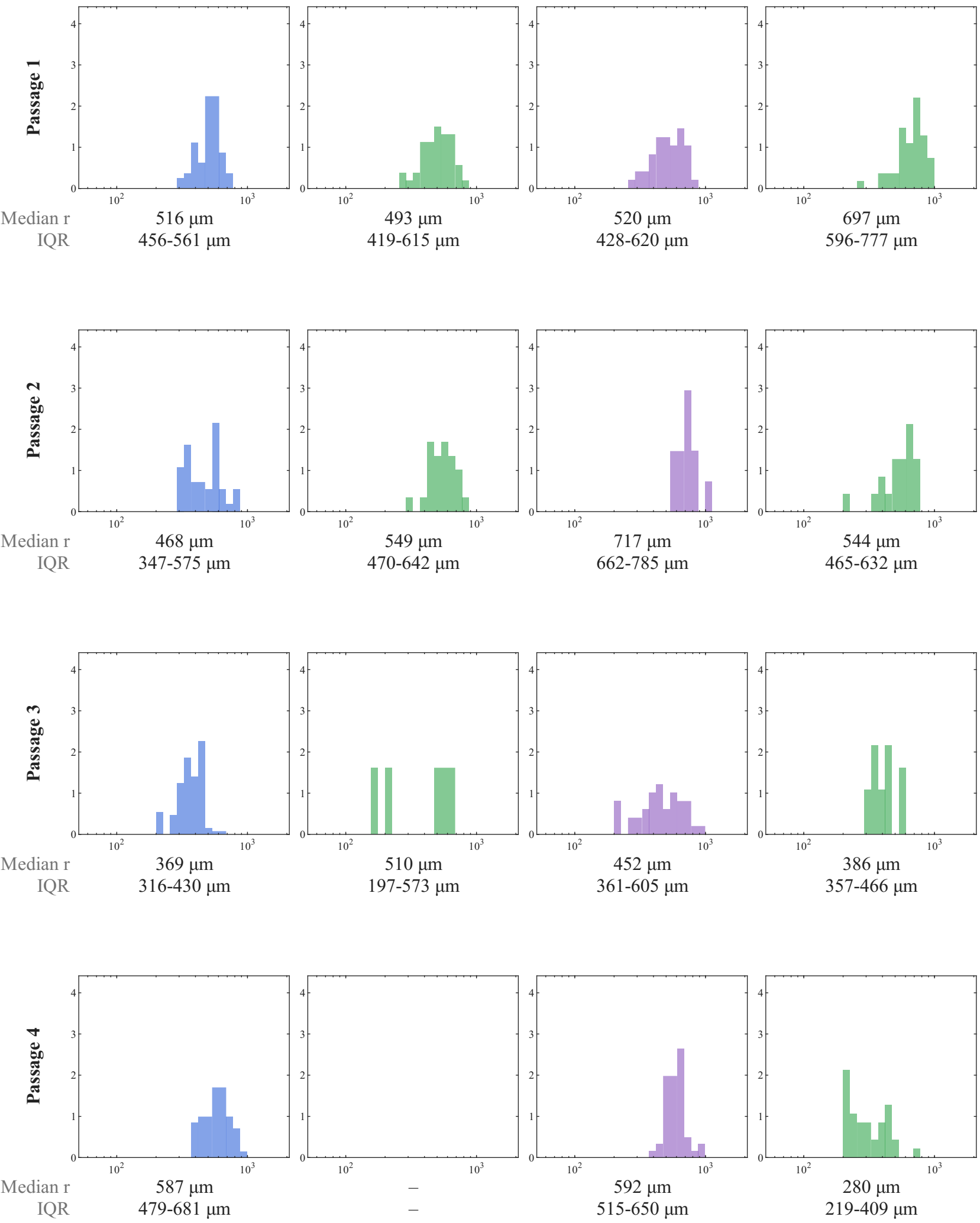
X axis: Colony Radius μm , Y axis: Density

Wild Type

est2 Δ

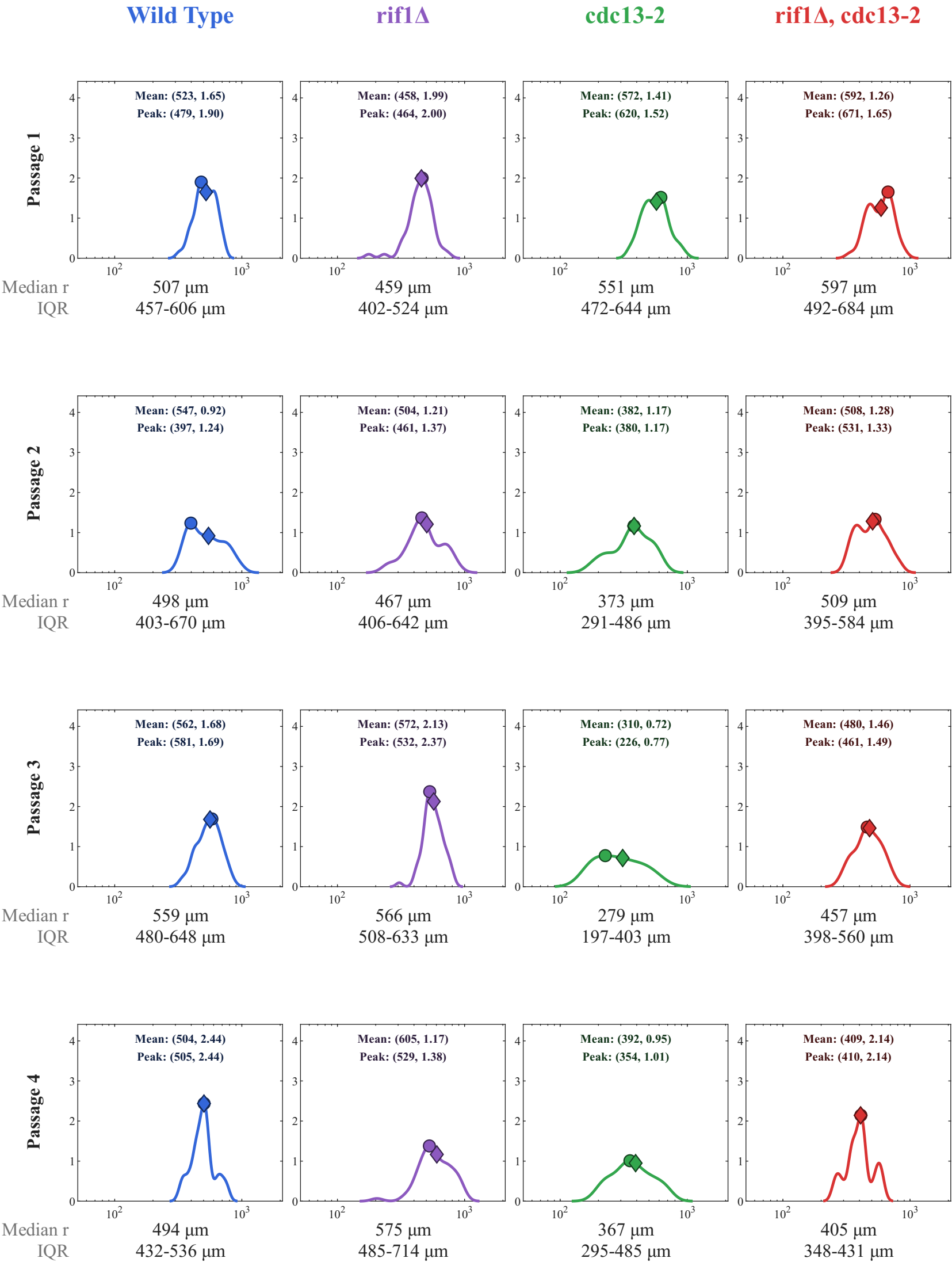
rif1 Δ

rif1 Δ , *est2* Δ



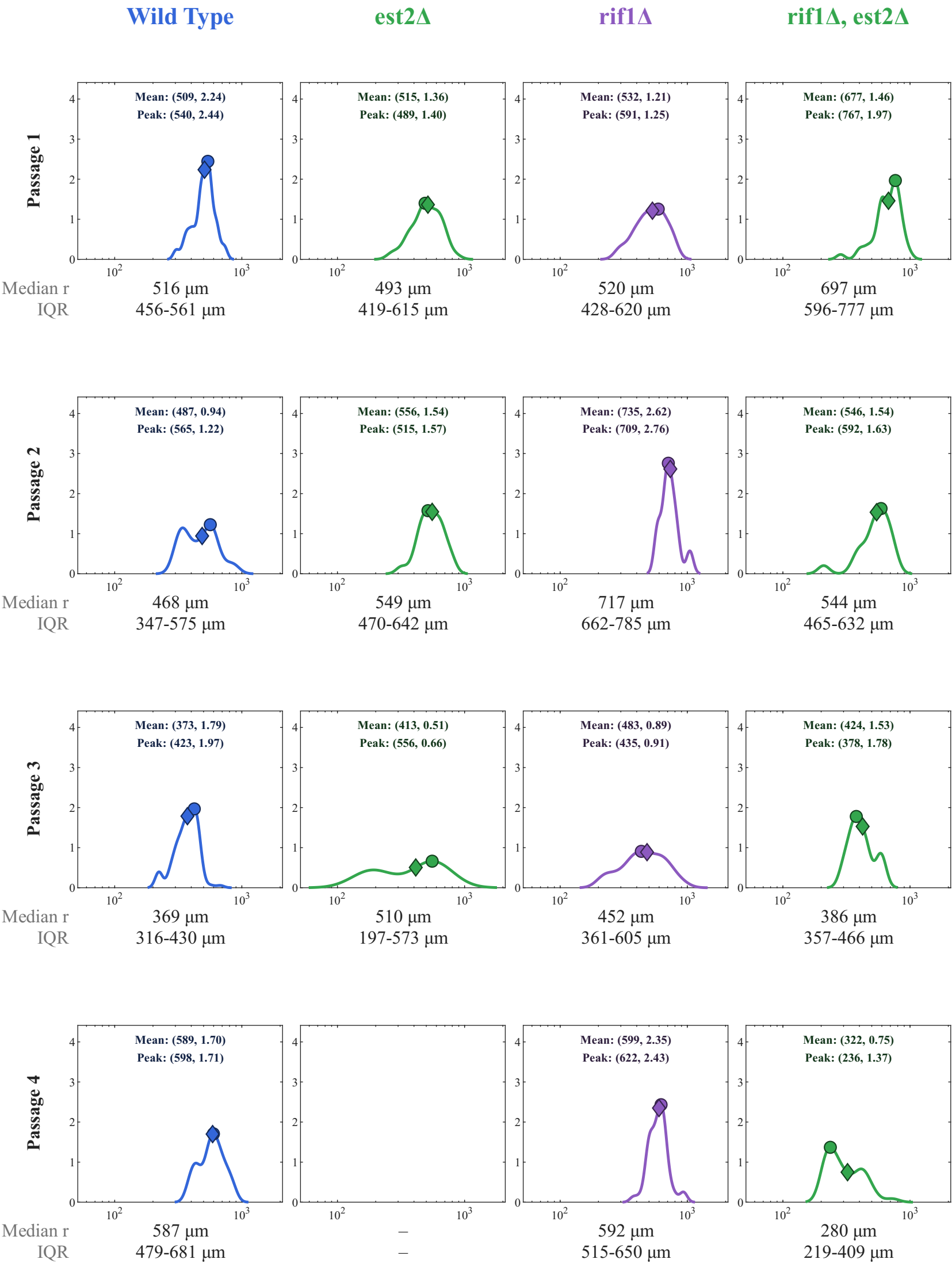
Kernel Density Estimates (Figure 2)

X axis: Colony Radius μm , Y axis: Density



Kernel Density Estimates (Figure 8)

X axis: Colony Radius μm , Y axis: Density



Wild Type

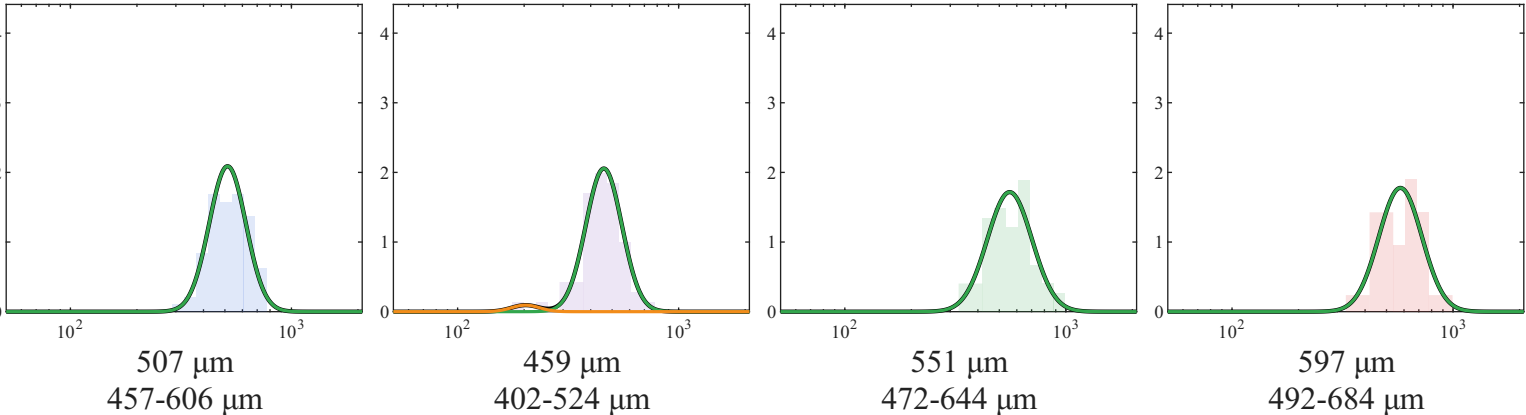
rif1 Δ

cdc13-2

rif1 Δ , *cdc13-2*

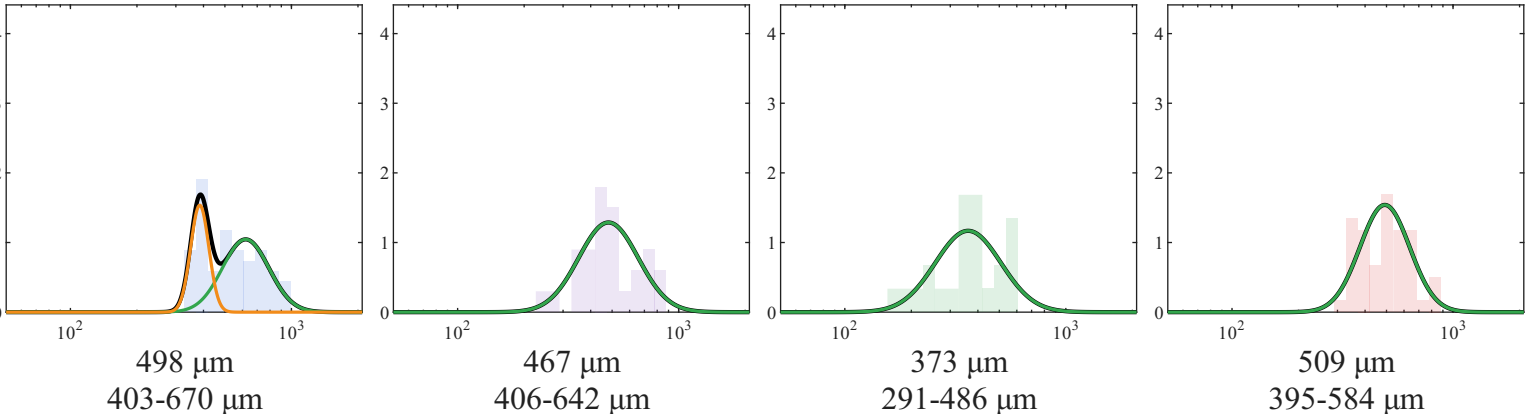
Passage 1

Median r
IQR



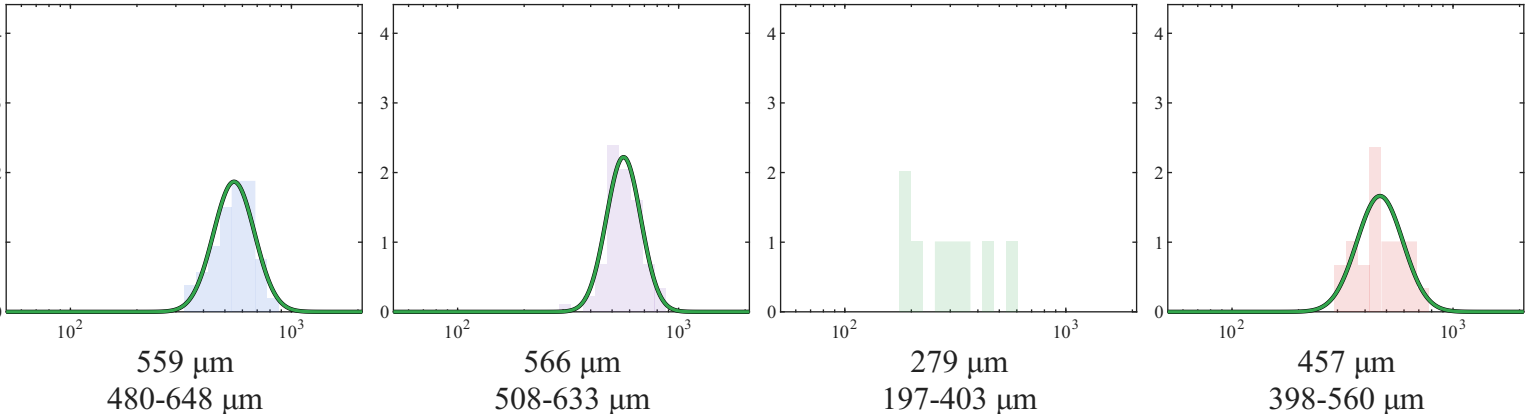
Passage 2

Median r
IQR



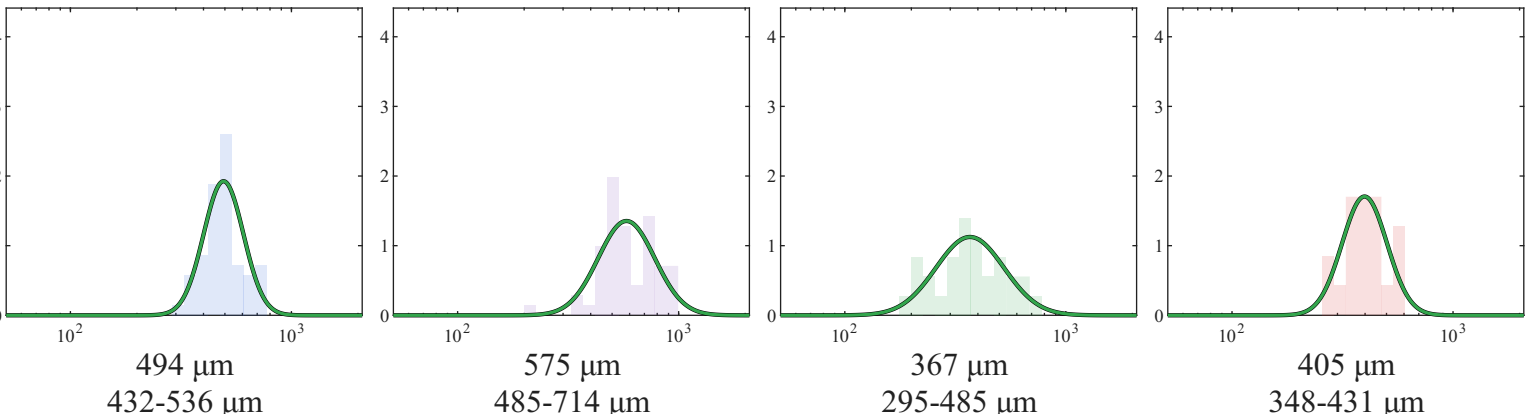
Passage 3

Median r
IQR



Passage 4

Median r
IQR

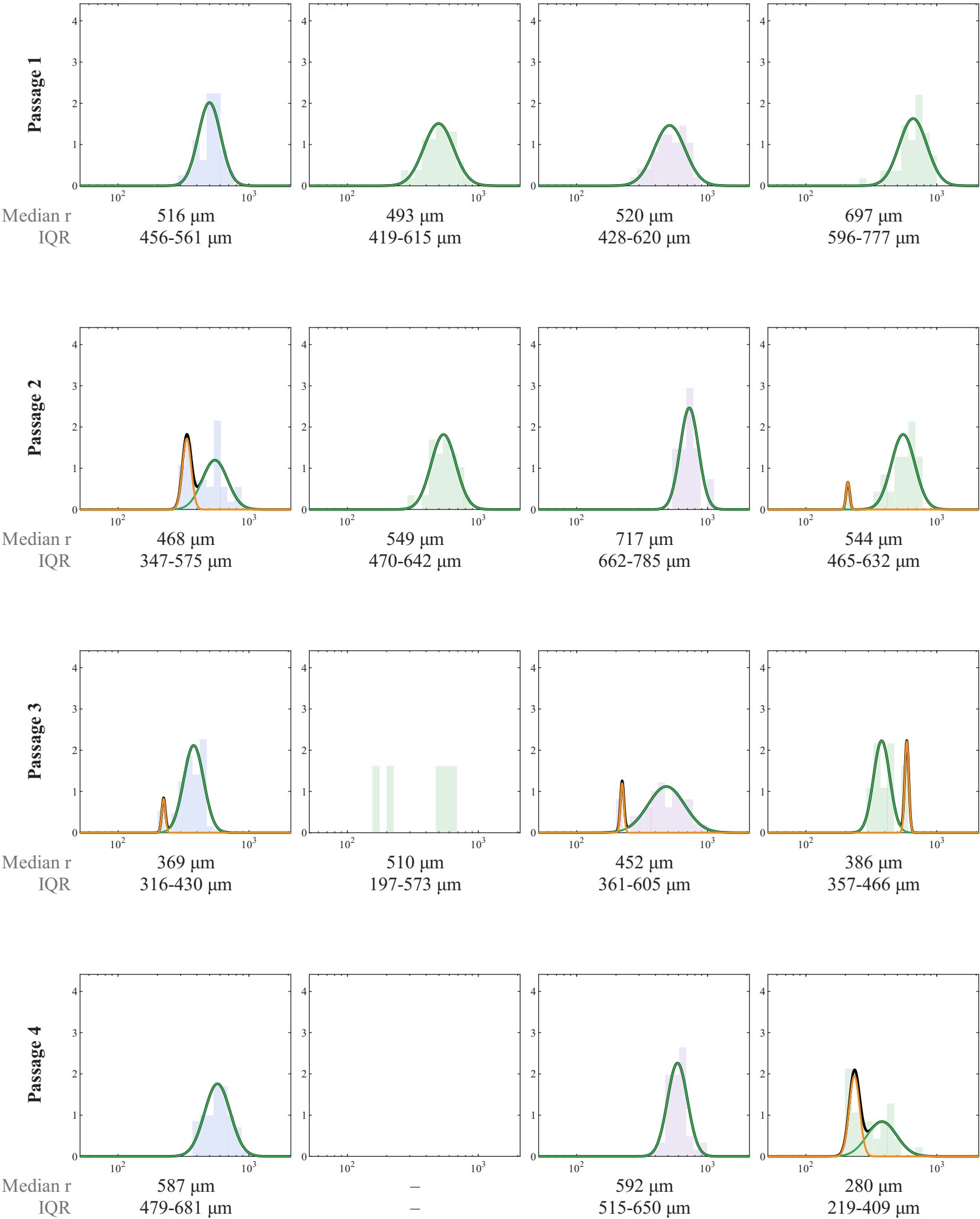


Wild Type

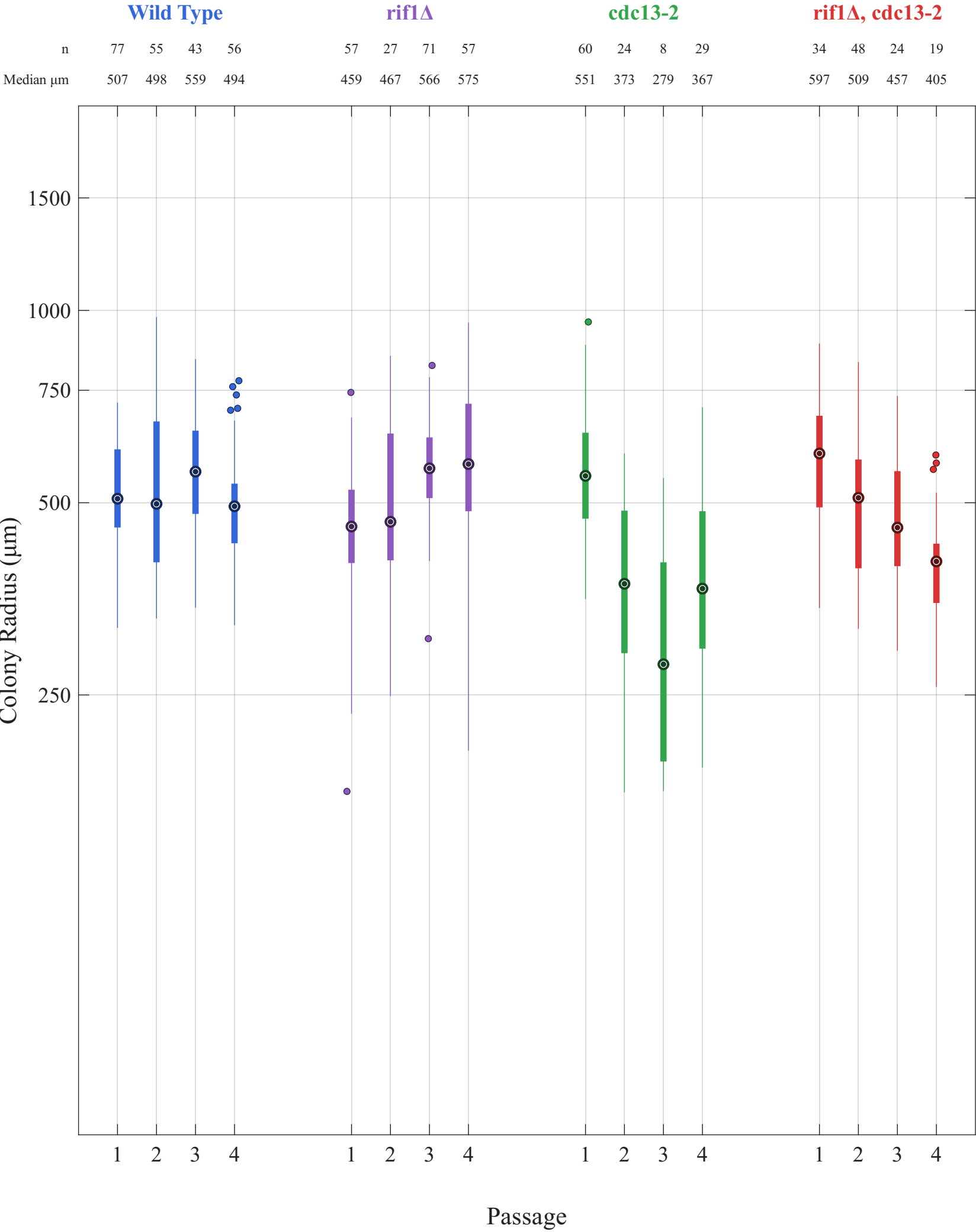
est2 Δ

rif1 Δ

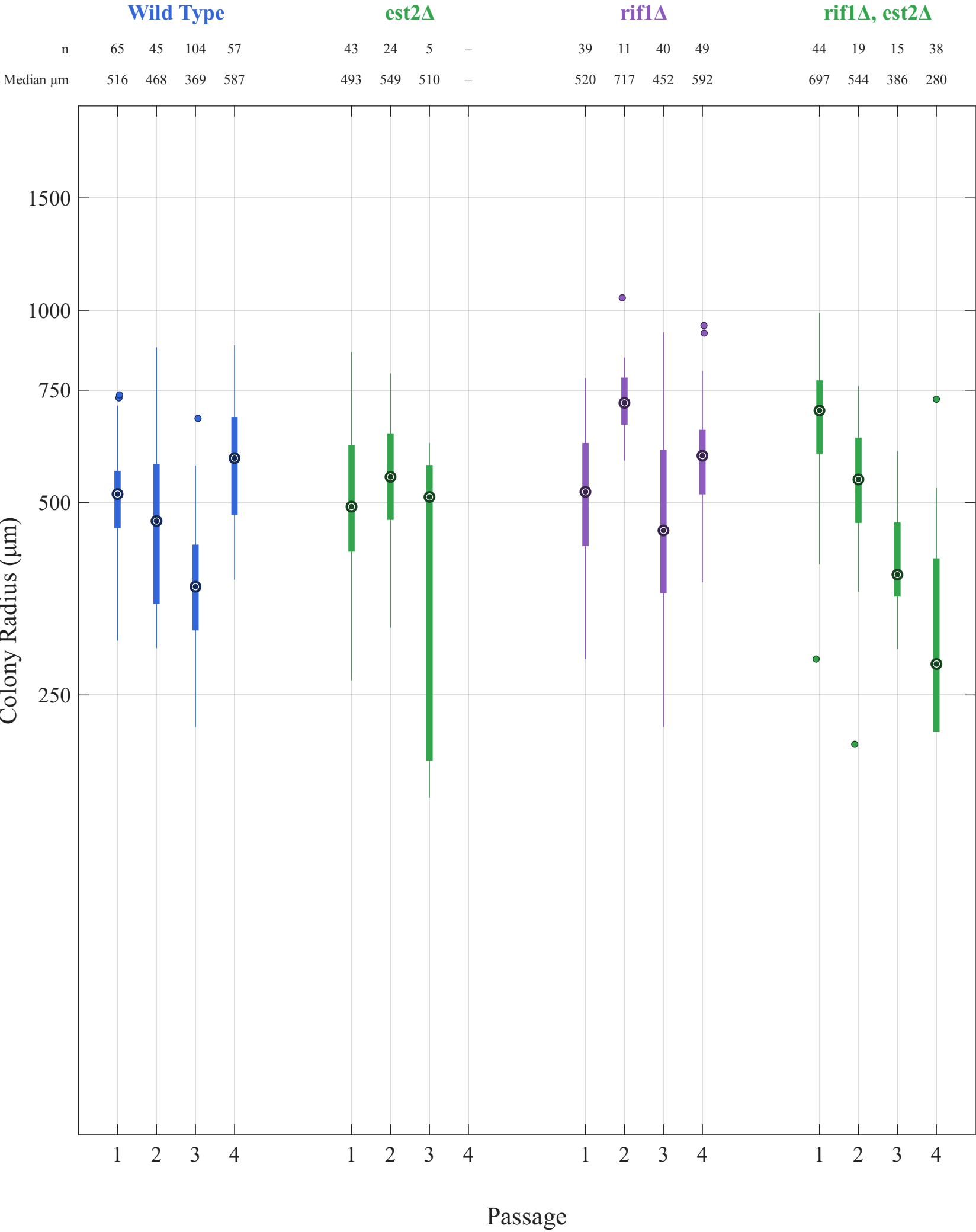
rif1 Δ , *est2* Δ



Colony Radius Distributions (Figure 2) X axis: Passage, Y axis: Colony Radius μm

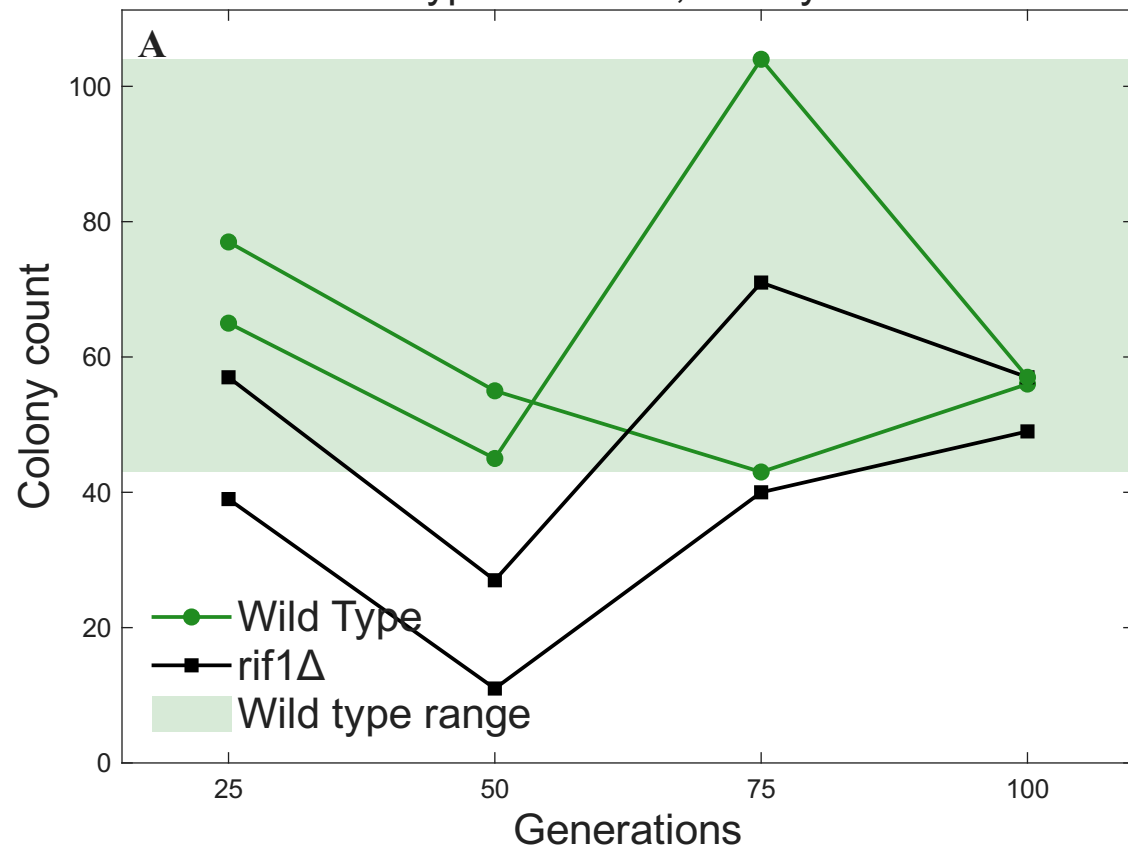


Colony Radius Distributions (Figure 8) X axis: Passage, Y axis: Colony Radius μm

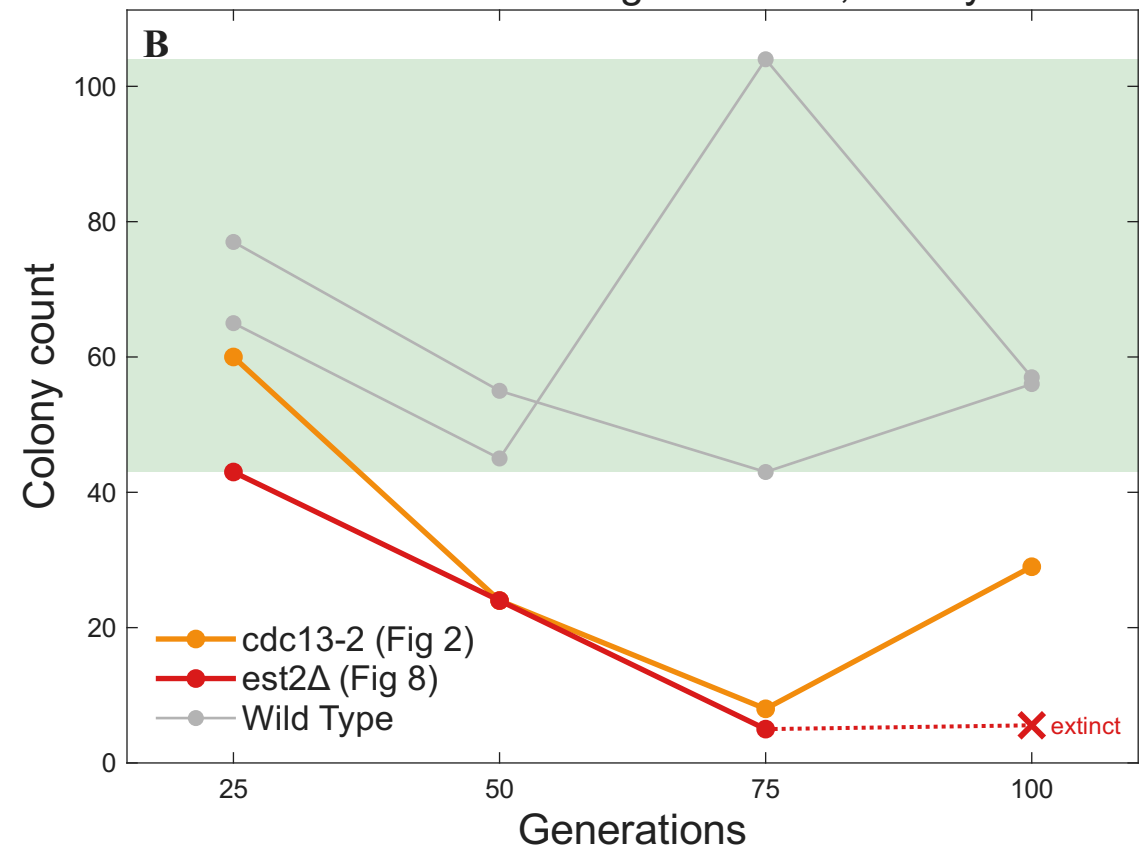


Senescent single mutants and non-senescent controls, measured values

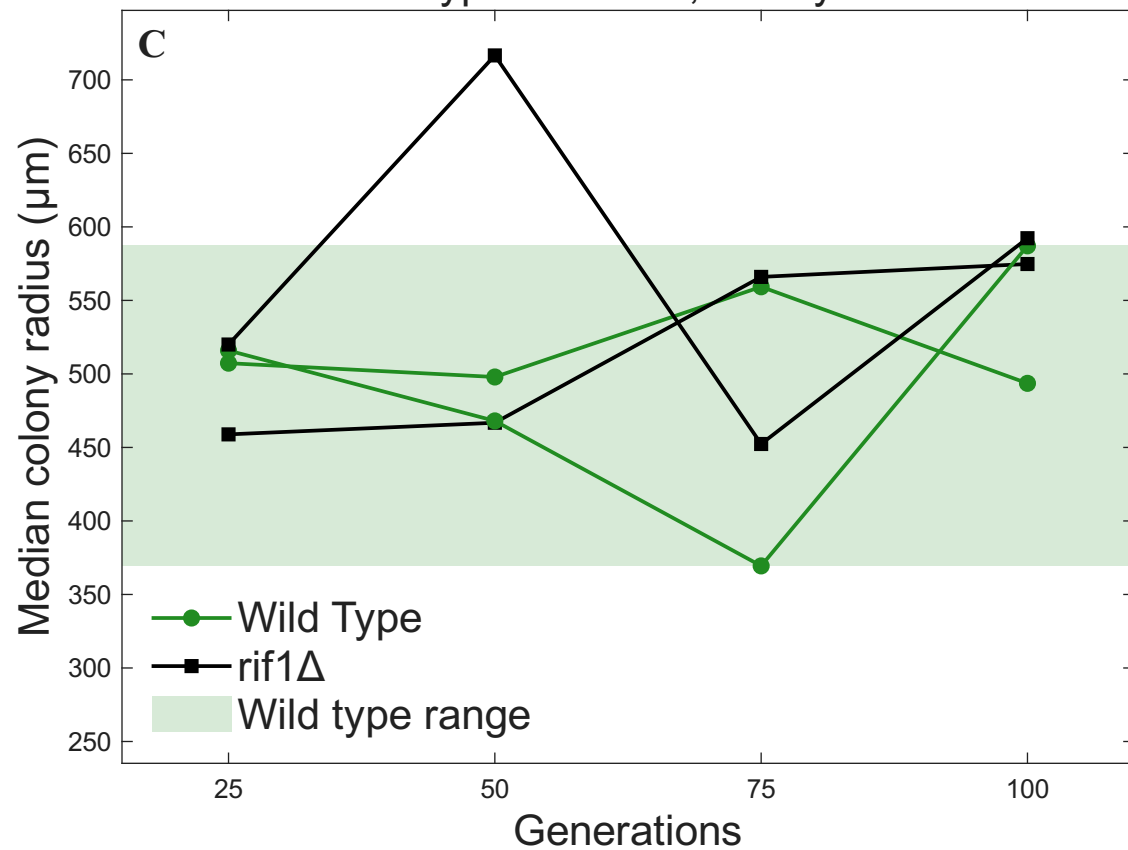
Wild type and rif1Δ, colony count



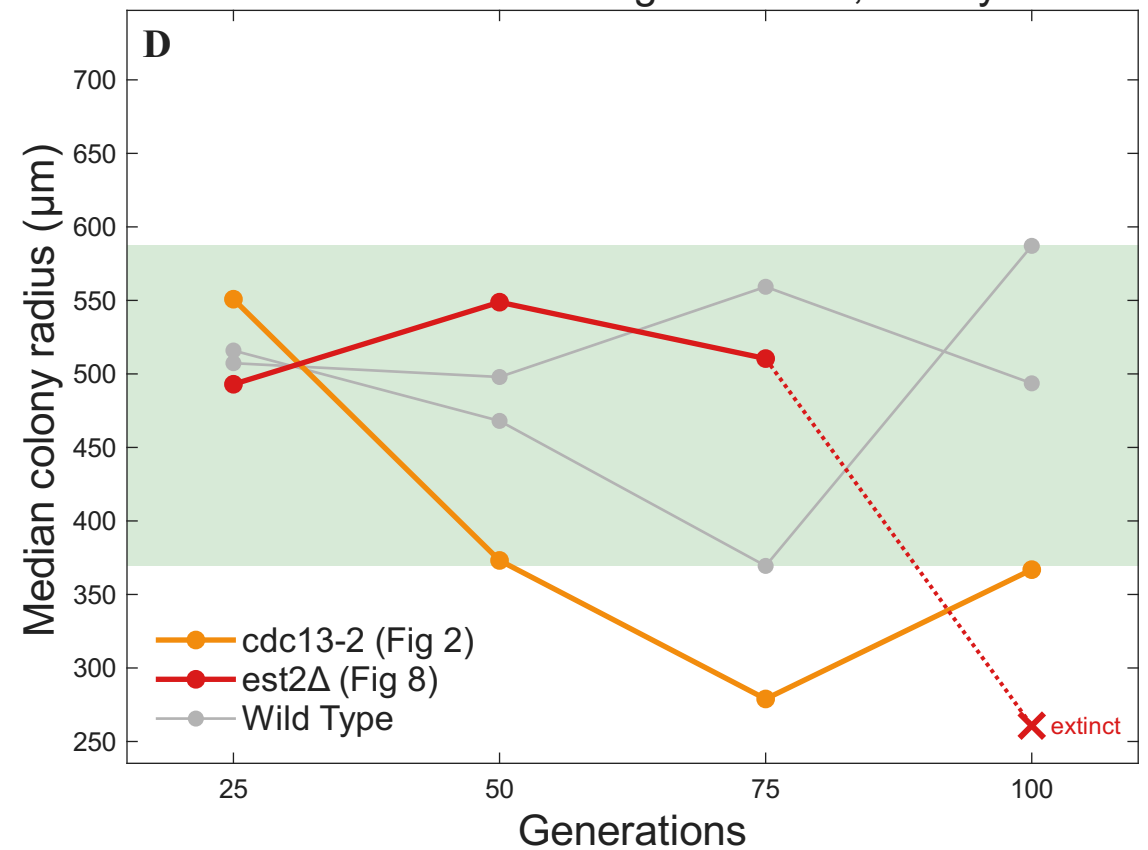
Telomerase-deficient single mutants, colony count



Wild type and rif1Δ, colony size

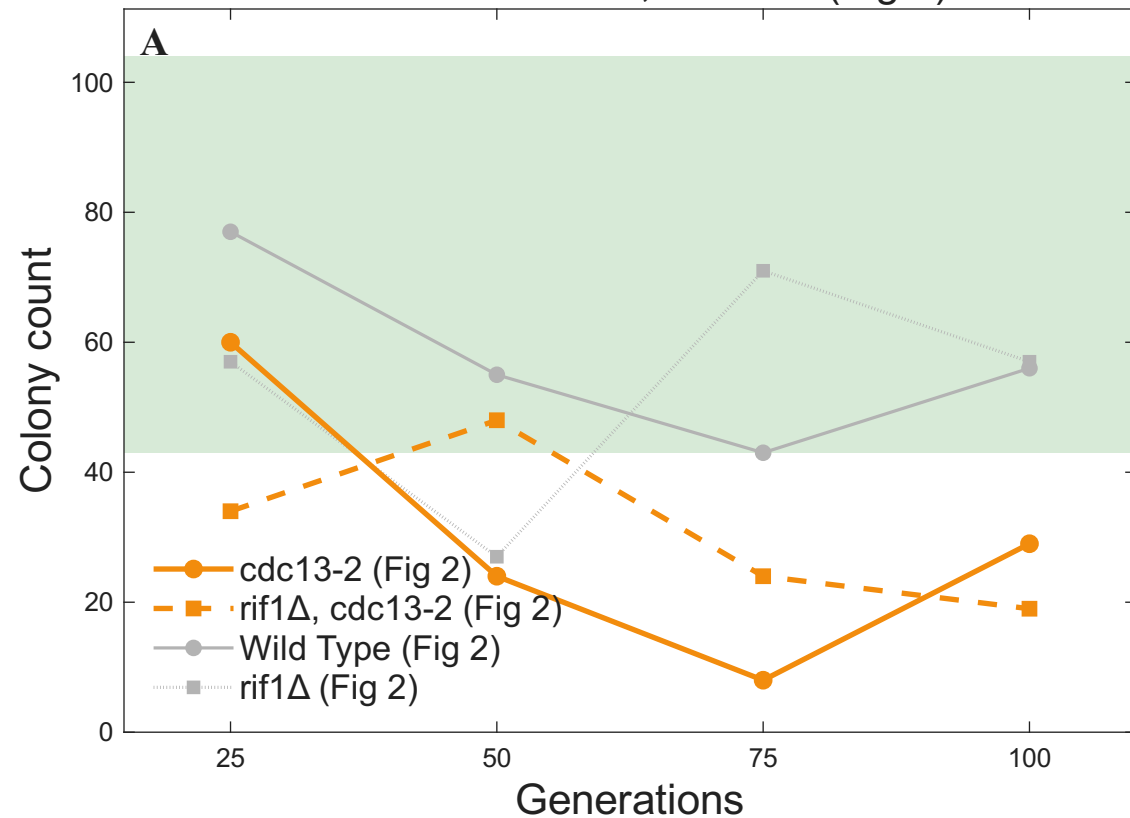


Telomerase-deficient single mutants, colony size

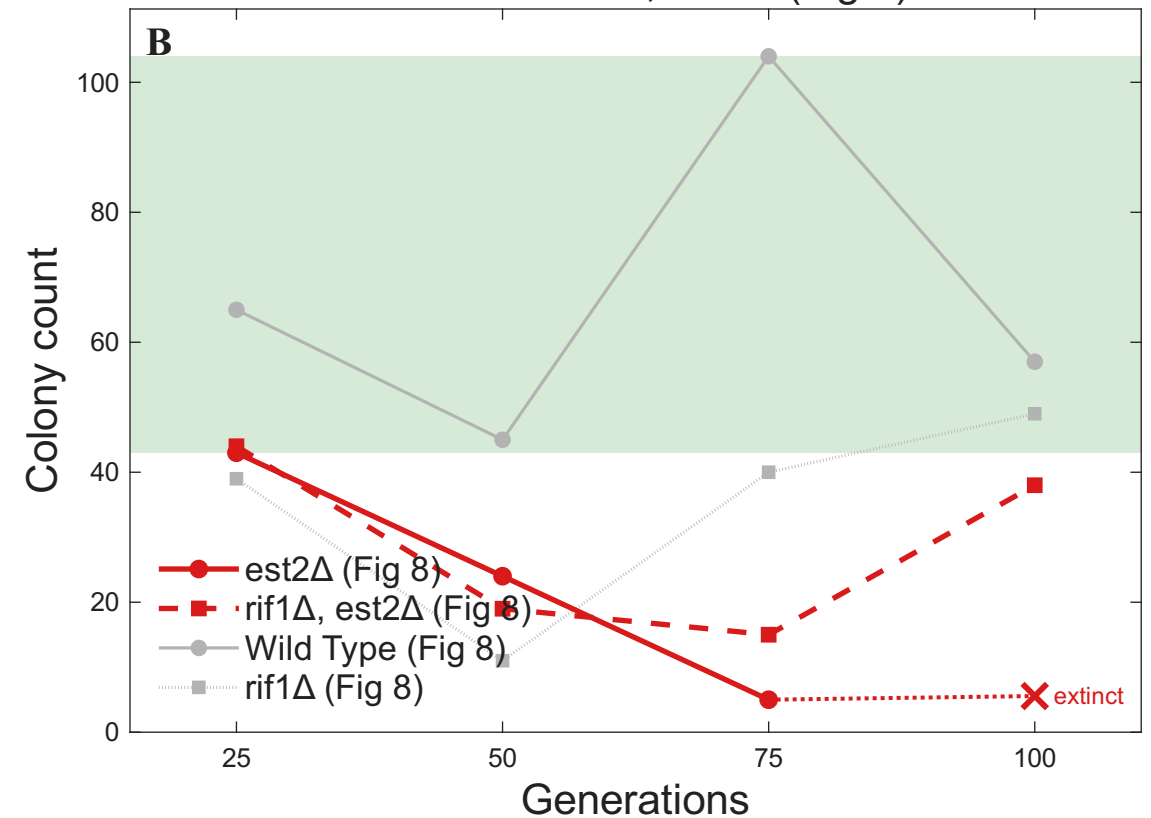


Senescent alleles, with and without rif1, measured values

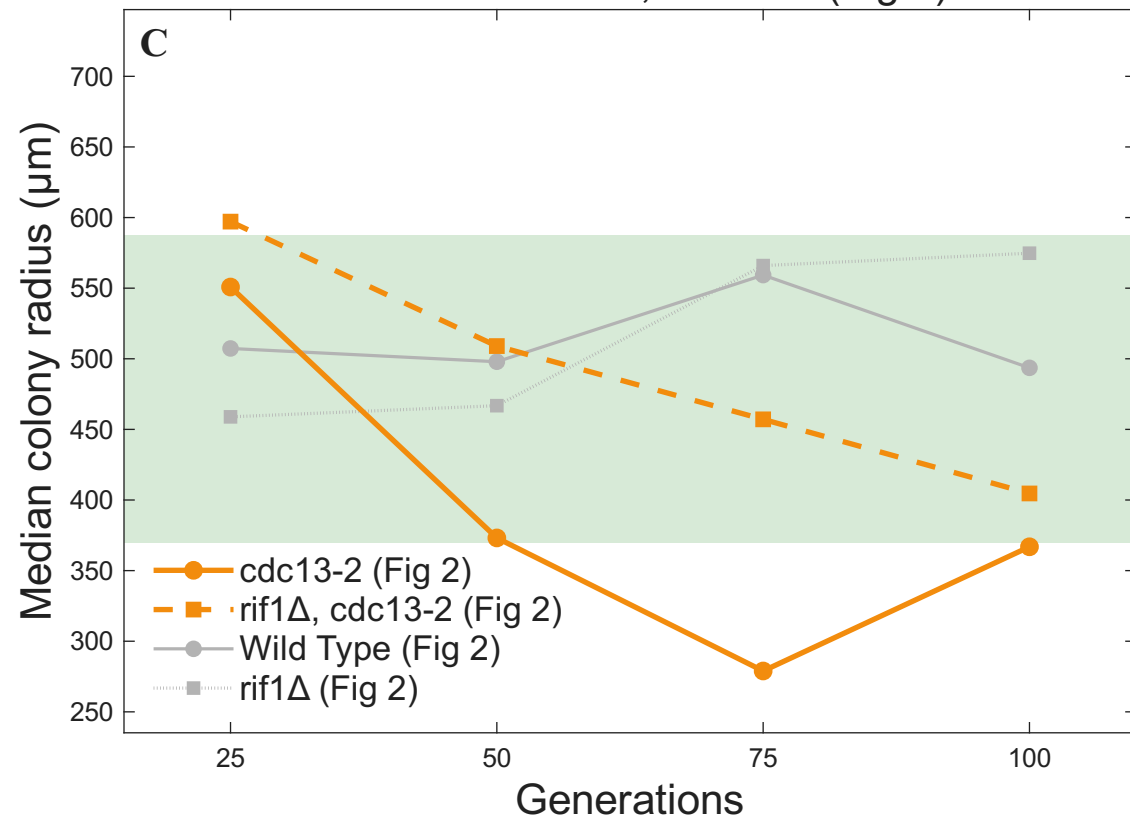
cdc13-2 vs rif1Δ, cdc13-2 (Fig 2)



est2Δ vs rif1Δ, est2Δ (Fig 8)



cdc13-2 vs rif1Δ, cdc13-2 (Fig 2)



est2Δ vs rif1Δ, est2Δ (Fig 8)

